Abstractive Text Summarization with Bidirectional LSTM on CNN/DailyMail Dataset

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Abstract

One of the main uses of LLM’s such as ChatGPT and Deepseek is for the summarization of long complex articles. Text summarizers can save casual readers valuable time and energy and even offer up ideas for users wishing to summarize text themselves. I'm aiming to create a text summarizer capable of paraphrasing news articles from the CNN/DailyMail summarization dataset. My project's goal is to utilize abstractive summarization using both an LSTM with attention and the BART Model then compares their performances. However with the failure of the LSTM model we can compare results without analytical results through a simple comparison of output.

Introduction

With the development of LLM’s such as chatGPT and Deepseek text summarization has become the norm for those looking to digest large amounts of information quickly. Many make the choice to toss the article into chatGPT for summarization rather than spending hours reading.

The performance of these LLM’s come from years of development and innovations. The transformer model has particularly had a huge impact on NLP and text summarization. My project aims to take things back to the times before transformers to test out LSTM models for summarization. This project details the performance of the BART transformer model and an Bi-LSTM with attention on the abstract summarization of the CNN/Dailymail news dataset.

Related Work

Text summarization has been explored using both abstractive and extractive methods. Early approaches focus on statistical techniques while recent work has moved toward deep learning focused approaches. Sirohi et al. (2021) surveyed several ML and LSTM based summarization approaches emphasizing the use of RNN's in modeling text sequences. On the same note, Kumar and Singh(2021) provided a review of summarization techniques showing the evolution of traditional algorithms to neural nets. Abel-Salam and Rafea (2022) demonstrated the use of transformer models such as BERT on extractive summarization tasks. This project adds to the literature by comparing the performance of Bi-LSTN with attention to the BART transformer for abstractive summarization of CNN/DailyMail.

Methods

In my implementation I began preparing the CNN/DailyMail data by importing from kaggle and utilizing pandas dataframes. The raw dataset contained 287,113 training points but due to computational limitations I decided to train my model on only 40,000. This dataset was transformed by splitting the articles and highlights on whitespace and mapping tokens to indexes in a vocabulary. To build vocabulary, token frequency was counted from the training set keeping tokens that showed up at least twice. Plus four special source tokens and three extra target tokens were added. A collate function then pads each batch of sequences so that every batch matches in shape.

My model is an encoder-decoder with attention. The encoder is a bidirectional LSTM that reads articles forward and backwards, producing context rich representations. The decoder is a unidirectional LSTM that at each step computes attention weights over encoder output, forms a context vector, then combines context with the previous word for a next word prediction. I initialize the decoder hidden state by linearly projecting the encoder's final forward and backwards states. During training I use teacher forcing 50 percent of the time feeding in the true prev token, or feeding in the models own prediction.

Then for each word in the target summary the decoder attends over the encoder outputs to pull in context. This produces next word probabilities, repeating until and end of sew token is emitted. I train my model for three epochs with a batch size of 8 using the adamW optimizer at a LR=0.005 and a cross entropy loss that ignores padding.

Results

After training the Bi-LSTM model I got a training loss of 6.5 and a validation loss of 7.0. Upon testing my model it was unable to produce viable summarizations of the news articles, outputting random characters or illegible text. For example a summary of an article involving a crime against young women resulted in the summary “police say found guilty to old year old year old year old was in the . . . .”. This shows that the model was unable to learn meaningful patterns in the data. This likely occurred because the model was not trained for long enough due to the lack of computational resources. Due to model failure I decided to skip the ROUGE evaluation knowing that it would be a waste to compare against the transformer model. To substitute this test I decided to implement the pretrained BART and compare its output to my LSTM. The BART model was loaded in from hugging face and the dataset was tokenized using the BART tokenizer. The BART model unsurprisingly performed perfectly on the news articles outputting accurate summarization of the articles.

Bi-LSTM

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<s> a gang of six men have been jailed for a total of 31 years after being convicted of a string of sexual offences against teenager girls. the

'police say found guilty to old year old year old year old was in the . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .'
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BART

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================================================================================
Original Article:
(CNN)The Palestinian Authority officially became the 123rd member of the International Criminal Court on Wednesday, a step that gives the court jurisdiction over alleged

================================================================================
Reference Summary (Ground Truth):
Membership gives the ICC jurisdiction over alleged crimes committed in Palestinian territories since last June .
Israel and the United States opposed the move, which could open the door to war crimes investigations against Israelis .

================================================================================
BART-Generated Summary:
The Palestinian Authority becomes the 123rd member of the International Criminal Court. The move gives the court jurisdiction over alleged crimes in Palestinian territori
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Conclusion

This project demonstrated the challenge of training BI-Directional LSTM models for abstractive summarization on large datasets such as CNN/DailyMail, particularly under limited computational resources. Despite a solid architecture the LSTM approach failed to generate meaningful summaries due to insufficient training time and data exposure. In contrast the pre-trained BART model showed impressive results. This reinforces the value of transformer models with large scale data and computational resources. These results showcase that when attempting projects or specific NLP tasks it may be more time efficient to choose pretrained models. This project has allowed me the opportunity to explore the creation of text summarization and has allowed me to create a Bi-directional LSTM from scratch. It also gave me the opportunity to learn more about attention and its implementation in NLP tasks.

Division of Labor

Justin Haysbert is responsible for all work presented in this project.

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